# BD-808 Bass Drum Module

Rev. 1

### Description

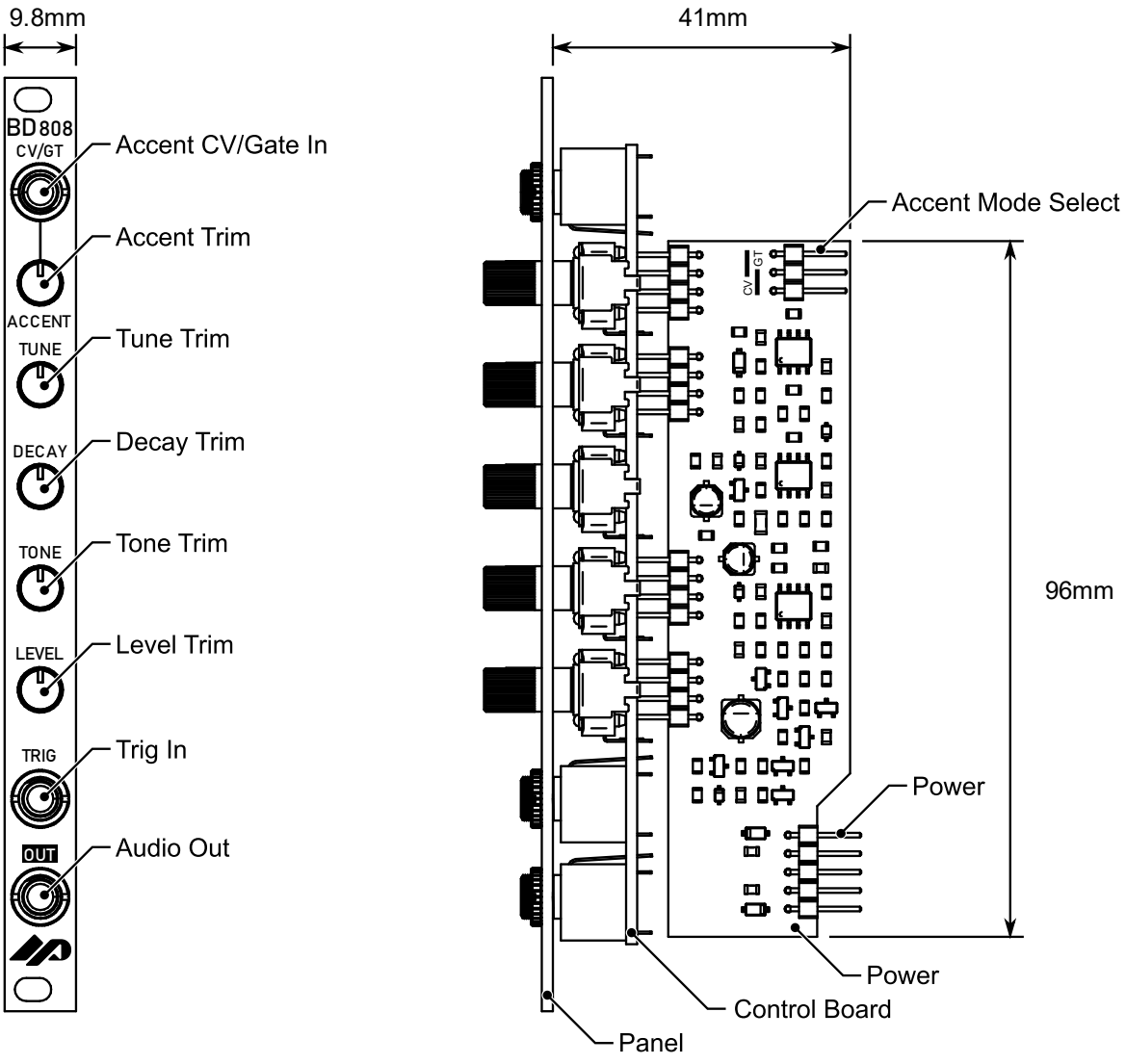
The BD-808 is a clone of the TR-808 Bass Drum, modified to fit the Eurorack size format and voltage requirements. It supports additional functionality not available in the original TR-808 drum machine, including features that make it more suited to the modular synthesizer format.

The BD-808 module fits in a compact 2hp form factor to minimize used rack space and has a max depth of 41 mm with recessed power header to fit almost any rack size. All inputs are robust with overvoltage and reverse-voltage protection.

### Features

- Compact 2hp form factor
- Internal gate to timed trigger conversion
- Dual selectable Accent (AC) input modes
    - AC Control Voltage (CV) mode
    - AC Gate (GT) mode
- Accent level trim
- Tune, Decay, Tone trim
- Input overvoltage and reverse-polarity protection
- Power input reverse-polarity protection
- Current Draw: +12V: 15mA, -12V: 12mA

## Module Layout



# Functionality

### Trig

The Trig input takes gate/trig inputs longer than 1ms in duration and converts them into 1ms timed pulses. This avoids any need for an external gate to trigger converter and prevents double-triggering on the falling edge of longer gate inputs. Note that input pulses shorter than 1ms are not altered. Trig inputs can be any voltage between 5V and 12V.

### Tune

The Tune trim sets the pitch of the drum sound. The pitch can range from a high 'tom' to a deep bass sound.

### Decay

The Decay trim allows the decay length of the drum sound to be lengthened or shortened. The decay circuit has been modified to enable longer decay times than a standard 808 can produce.

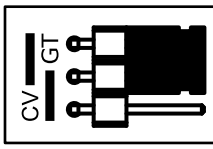
### Tone

The Tone trim adjusts the output filter of the drum to produce a snappier kick sound or a smoother bass sound.

### Accent

The Accent (AC) effect adds a dynamic feel to a drum sequence by modifying the strength of the drum hit (i.e. volume/tone). The Accent effect can be set by the Accent trim alone (with no physical inputs to the AC CV/GT jack), or with external voltages. The BD-808 has two accent modes for external voltages, selectable with a jumper and 3-pin mode-select header: AC CV Mode and AC GT Mode.

*AC CV Mode*

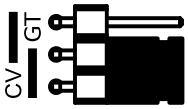



*AC CV Mode Jumper Position*

AC CV Mode allows the Accent setting to be modified by an external analog CV input. This input can range from 0V to 6V, with 0V being a 'light' drum hit and 6V being a 'hard' hit. CV input voltages larger than 6V will not damage the module, however they will also not further increase the accent setting.

In AC CV mode the Accent trim acts as an offset for the CV input. The Accent trim setting will always be active and applied to every beat, regardless of the presence of a CV voltage.

*AC GT Mode*



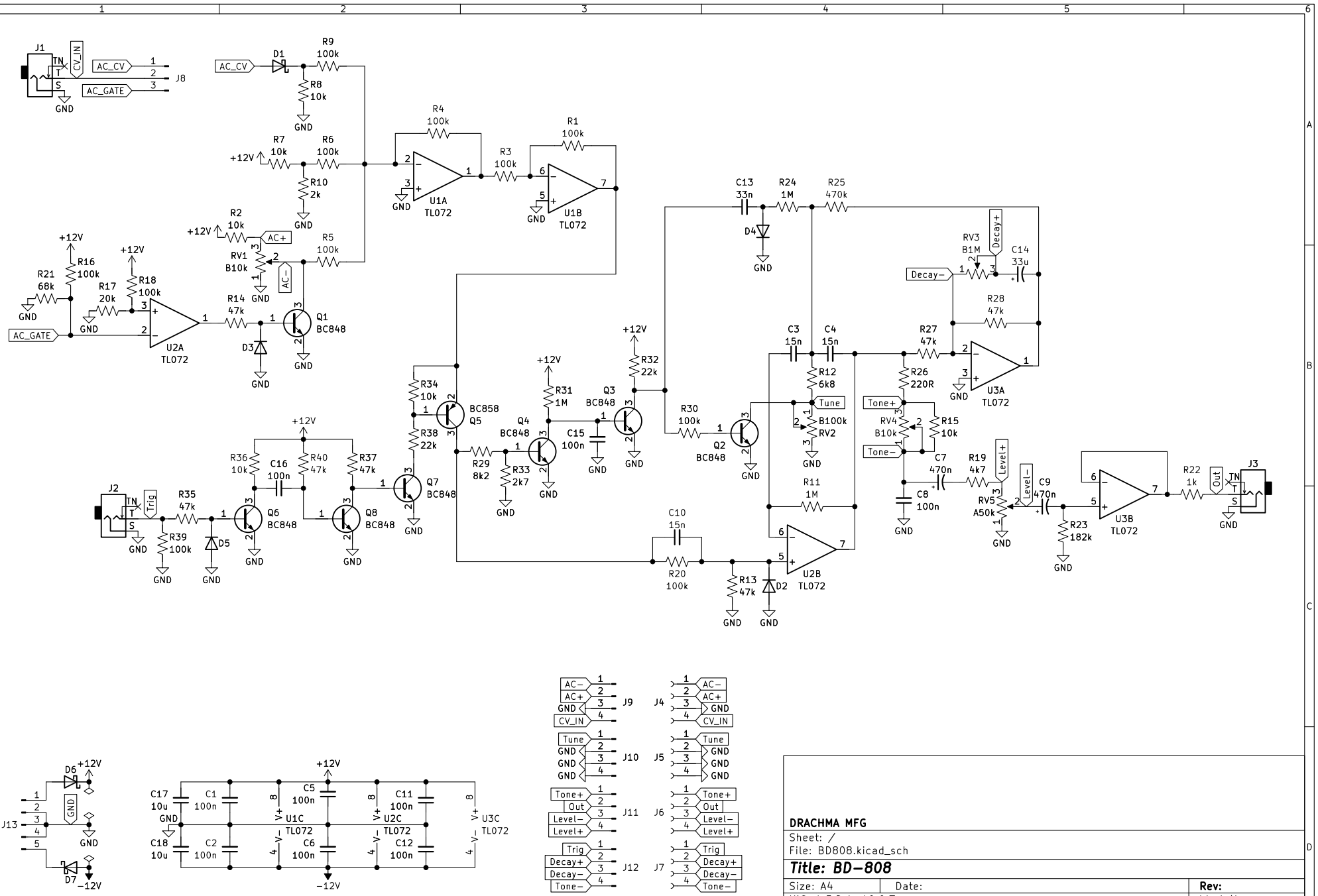

*AC GT Mode Jumper Position*

AC GT mode allows the Accent effect to be gated (On/Off) rather than continuously variable, similar to the functionality of the original TR-808. To gate the Accent effect (i.e., turn it On), any voltage input between 5V and 12V (High) can be used. A voltage input less than 2V (Low) will turn the Accent effect Off.

Though the AC GT input should be treated like any other digital gate input, note that it is internally pulled up to +5V. Floating input signals will be treated as a High input, allowing for the use of the Accent trim with no physical inputs to the AC CV/GT jack. In order to leave beats un-accented in AC GT Mode, the AC GT input must be pulled down to G

## Bill of Materials

| Value | Qty. | Name on Board | Description | Footprint |
|---|---|---|---|---|
| 220Ω | 1 | R26 | 220Ω 1% 0.125W Resistor | 0805 |
| 1kΩ | 1 | R22 | 1kΩ 1% 0.125W Resistor | 0805 |
| 2kΩ | 1 | R10 | 2kΩ 1% 0.125W Resistor | 0805 |
| 2.7kΩ | 1 | R33 | 2.7kΩ 1% 0.125W Resistor | 0805 |
| 4.7kΩ | 1 | R19 | 4.7kΩ 1% 0.125W Resistor | 0805 |
| 6.8kΩ | 1 | R12 | 6.8kΩ 1% 0.125W Resistor | 0805 |
| 8.2kΩ | 1 | R29 | 8.2kΩ 1% 0.125W Resistor | 0805 |
| 10kΩ | 6 | R2, R7, R8, R15, R34, R36 | 10kΩ 1% 0.125W Resistor | 0805 |
| 20kΩ | 1 | R17 | 20kΩ 1% 0.125W Resistor | 0805 |
| 22kΩ | 2 | R32, R38 | 22kΩ 1% 0.125W Resistor | 0805 |
| 47kΩ | 7 | R13, R14, R27, R28, R35, R37, R40 | 47kΩ 1% 0.125W Resistor | 0805 |
| 68kΩ | 1 | R21 | 68kΩ 1% 0.125W Resistor | 0805 |
| 100kΩ | 11 | R1, R3, R4, R5, R6, R9, R16, R18, R20, R30, R39 | 100kΩ 1% 0.125W Resistor | 0805 |
| 182kΩ | 1 | R23 | 182kΩ 1% 0.125W Resistor | 0805 |
| 470kΩ | 1 | R25 | 470kΩ 1% 0.125W Resistor | 0805 |
| 1MΩ | 3 | R11, R24, R31 | 1MΩ 1% 0.125W Resistor | 0805 |
| 15nF | 3 | C3, C4, C10 | 15nF 50V C0G MLCC | 0805 |
| 33nF | 1 | C13 | 33nF 50V C0G MLCC | 0805 |
| 100nF | 1 | C8 | 100nF 50V C0G MLCC | 1206 |
| 100nF | 8 | C1, C2, C5, C6, C11, C12, C15, C16 | 100nF 50V X7R MLCC | 0805 |
| 470nF | 2 | C7, C9 | 470nF 50V Elec. Cap. | 4x5.4mm SMD |
| 10uF | 2 | C17, C18 | 10uF 50V X5R MLCC | 0805 |
| 33uF | 1 | C14 | 33uF 25V Elec. Cap. | 5x5.7mm SMD |
| 1N4148 Diode | 4 | D2, D3, D4, D5 | 1N4148 Gen. Purp. Diode | SOD-323 |
| Schottky Diode | 3 | D1, D6, D7 | Schottky Diode 40V | SOD-123 |
| BC846 NPN | 7 | Q1, Q2, Q3, Q4, Q6, Q7, Q8 | BC846 NPN Transistor | SOT-23 |
| BC856 PNP | 1 | Q5 | BC856 PNP Transistor | SOT-23 |
| TL072 | 3 | U1, U2, U3 | TL072 Dual Op Amp | SOIC-8 150mil |
| B10k Pot Lin | 2 | RV1, RV4 | 10k Lin. Potentiometer | Alpha 9mm vertical |
| B100k Pot Lin | 1 | RV2 | 100k Lin. Potentiometer | Alpha 9mm vertical |
| B1M Pot Lin | 1 | RV3 | 1M Lin. Potentiometer | Alpha 9mm vertical |
| A50K Pot Log | 1 | RV5 | 50k Log. Potentiometer | Alpha 9mm vertical |
| 3.5mm Jack | 3 | J1, J2, J3 | 3.5mm Mono Jack | PJ-3001F |
| 5-Pin Header | 1 | J13 | 2.54mm Header 5x1 | 2.54mm 5-pin right angle |
| 4-Pin Header | 4 | J4/J9, J5/J10, J6/J11, J7/J12 | 2.54mm Header 4x1 | 2.54mm 4-pin right angle |
| 3-Pin Header | 1 | J8 | 2.54mm Header 3x1 | 2.54mm 3-pin right angle |
| 2-Pos. Jumper | 1 | N/A (For use on J8) | 2.54mm 2-Pos. Jumper | N/A |




**DRACHMA MFG**

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| KiCad E.D.A. 10.0.7 | | Id: 1/1 |